

LONG EXAM – Information Security

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Course: INF233

1. Effectiveness of the AI-enabled IDPS

The AI-enabled Intrusion Detection and Prevention System (IDPS) demonstrates a moderate level of effectiveness in identifying and mitigating cyber threats within enterprise environments. Through the application of machine learning and artificial intelligence, the system can analyze large volumes of network traffic and detect suspicious activities more efficiently than traditional rule-based systems. However, despite its advantages, the IDPS still encounters challenges in recognizing sophisticated attacks such as Advanced Persistent Threats (APTs) and zero-day exploits. In addition, the occurrence of false positives increases the workload of Security Operations Center (SOC) analysts and may contribute to alert fatigue. Studies in cybersecurity emphasize that while AI-based systems improve detection accuracy, continuous model training, high-quality datasets, and human oversight remain essential to ensure reliable security operations in real-world environments.

2. Effect of Dataset Representativeness

Dataset representativeness significantly affects the performance and reliability of AI-driven intrusion detection systems. A dataset that accurately reflects real-world network traffic allows the system to generalize effectively and identify both known and emerging cyber threats. When datasets are incomplete, outdated, or biased, the system may fail to detect certain attacks and may also generate inaccurate classifications. According to academic research in artificial intelligence and cybersecurity, diverse and balanced datasets are essential in training robust machine learning models. Therefore, organizations must ensure the use of realistic and continuously updated datasets to improve the effectiveness and adaptability of intrusion detection systems in dynamic enterprise environments.

3. Role of Evaluation Rigor

Evaluation rigor plays an important role in determining the validity and credibility of an AI-based intrusion detection system. Researchers and cybersecurity professionals commonly use metrics such as precision, recall, accuracy, and F1-score to measure system performance. Rigorous evaluation methods help determine whether the system can consistently perform under different conditions and detect attacks accurately in imbalanced datasets. Inconsistent or weak evaluation practices may lead to misleading results and reduce confidence in the system's effectiveness. Therefore, standardized testing procedures, proper benchmarking, and continuous performance assessments are necessary to ensure the reliability and practical applicability of intrusion detection systems.

4. Influence of Model Robustness and Deployment Readiness

Model robustness and deployment readiness are essential factors that influence the practical success of AI-enabled intrusion detection systems. Model robustness refers to the

ability of the system to maintain stable performance despite variations in network traffic, cyberattack patterns, and environmental conditions. Deployment readiness, on the other hand, focuses on the system's compatibility with existing enterprise infrastructure such as Security Information and Event Management (SIEM) platforms. Research studies highlight that even highly accurate AI models may fail in real-world implementation if they cannot adapt to dynamic environments or integrate efficiently with operational tools. Consequently, organizations must prioritize both technical reliability and operational compatibility to maximize the value of AI-based security solutions.

5. Impact of Explainability and Trust

Explainability and trust are critical factors that influence the adoption and effectiveness of AI-based cybersecurity systems. Explainability refers to the system's ability to provide understandable reasons behind its decisions and generated alerts. When analysts can clearly interpret how the AI reached a conclusion, they are more likely to trust the system and respond effectively to threats. However, many AI models operate as "black box" systems, where decision-making processes are difficult to understand. This lack of transparency can reduce analyst confidence and delay response times. Academic studies suggest that improving explainability through interpretable AI models and visualization tools can strengthen trust, improve decision-making, and enhance overall cybersecurity performance.

References

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